

Juan David Munoz Bolanos

Junior Optical System Engineer

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Summary

Junior system engineer with hands-on lab work building, aligning, and calibrating optical laser systems using spatial light modulators and holography. I'm comfortable with PC and control technology, used to digging into technical problems systematically.

Research/Professional Experience

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Optical Systems Engineer / PhD Researcher

Optomechatronic System Design: Designed and built a multiphoton microscope and an automated wavefront sensing system from scratch, integrating lasers, spatial light modulators, and deformable mirrors.

Data Analysis & Algorithms: Developed Python and C++ pipelines for real time signal processing, iterative phase retrieval, and wavefield simulations to optimize system performance.

Systems Engineering & Quality: Applied V Model methodologies, Design of Experiments, and FMEA to track requirements and systematically resolve technical defects during prototype iterations.

Lab Management & Safety: Managed a high power laser laboratory, ensuring EH&S compliance, maintaining equipment logs, and coordinating component sourcing with suppliers like Thorlabs and Edmund Optics.

Aug 2018 to Oct 2019 **INTECOL S.A.S.** *Medellín, Colombia*
Junior Machine Vision Engineer (Field Service)

Machine Vision Pipelines: Engineered and deployed Halcon and Cognex VisionPro image processing pipelines for high speed industrial production lines, improving quality control.

Illumination Design: Designed custom bright field and dark field illumination systems to reliably detect defects, liquid levels, and sealing states on fast moving products.

Industrial Automation: Integrated vision systems with robotic arms and PLC controllers, executing on site testing, validation, and customer sign off.

Technical Competencies

Software & Tools

Python C++ LabVIEW Halcon OpenCV ZEMAX MATLAB Git

Optical Systems

Adaptive Optics Wavefront Sensing Microscopy Spectroscopy Laser Systems
DMDs

Systems & Quality

V-Model 8D Problem Solving Design of Experiments ISO 9001

Education

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Doctor of Philosophy in Optical Metrology

Oct 2019 to Sep 2021	Friedrich Schiller Univ. Jena <i>Master of Science in Photonics</i>	Jena, Germany
Aug 2012 to Aug 2018	University of Cauca <i>Bachelor of Science in Physics Engineering</i>	Popayán, Colombia

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

- Biking/Hiking
- Bachata Dancing
- Harmonica
- Learning German

Languages

- English (Fluent)
- German (Intermediate B2)
- Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte <i>Medical University of Innsbruck</i> monika.ritsch-marte@i-med.ac.at	PD Dr. Christoph Krafft <i>Leibniz IPHT</i> christoph.krafft@leibniz-ipht.de
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